Measuring H_0 with Gravitational-Wave Standard Sirens

Elisa Genot s5532930
Letícia Sá Arruda s5717965
Michael Jacob Pineda s5944163
Horia-Emilian Nita s5756758
Elliot Craig s5617863

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Abstract

Determining the Hubble constant (H_0) is a key challenge in cosmology today, particularly due to disagreements between results obtained by different methods. In this report the novel Standard Siren method, which combines both gravitational wave and electromagnetic signals, is implemented. It reproduces the results obtained by a paper by B.P.Abbott et al. [1] which analyses data from the gravitational wave GW170817 in combination with the γ-ray burst GRB 170817A. The approach taken in this report is based on a Bayesian statistical analysis using posterior samples on the luminosity distance as a likelihood function by implementing Kernel density estimation (KDE). The value for H_0 is obtained using an MCMC algorithm to explore the parameter space spanned by the cosine of the inclination angle ($\cos(\iota)$) and H_0. Corrections due to uncertainties in the redshift as well as peculiar velocity corrections are applied. The value obtained is $H_0 = 68^{+13}_{-6}$ km s^{-1} Mpc^{-1} which is comparable to the value of $H_0 = 70.0^{+12.0}_{-8.0}$ km s^{-1} Mpc^{-1} found by B.P.Abbott et al. [1].

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1 Introduction

The calculation of the Hubble constant H_0, representing the current expansion rate of the universe, is a central challenge in cosmology [2]. For nearly a century, scientists have improved on Edwin Hubble's 1929 estimate of the constant $H_0 = 500$ km s^{-1} Mpc^{-1} [2] [3]. However, there is a fundamental disagreement in the measured values of H_0 coined the "Hubble tension" [2] [4] [5]. There is a statistical disagreement between two main methods of determining H_0: the early universe measurements that infer H_0 from the cosmic microwave background (CMB) and assume the standard cosmological model obtain $H_0 = (67.27 \pm 0.60)$ km s^{-1} Mpc^{-1}, the late universe measurements based on empirical, local measurements independent of the standard cosmological model obtain $H_0 = (73.30 \pm 1.04)$ km s^{-1} Mpc^{-1} [4] [6] [7]. The reasons behind this discrepancy are still unclear, they could be due to unrecognized systematic errors or flaws in the standard cosmological model, thus it is imperative to find new independent methods of measuring H_0 [4].

In particular, the standard siren method in which gravitational wave signals give direct measurements of the luminosity distance, could play a central role in resolving the Hubble tension [1] [8] [4]. For this a binary neutron star (BNS) inspiral could be used as it is predicted to create detectable gravitational-waves before collision [9]. On August 17 2017 the first binary neutron star merger was observed by the Advanced LIGO and Advanced Virgo gravitational-wave detectors [9]. This measurement involved two distinct but related signals, the gravitational wave (GW) GW170817, and a short γ-ray burst (GRB) GRB 170817A, 1.7 seconds after the collision [9]. These measurements seem to confirm the theoretical predictions that short GRBs are caused by BNS mergers [9]. From this a better understanding of key cosmological processes can be gained, such as calculating the Hubble constant H_0 [9] [1].

The aim of this project is twofold; reproducing Figure 2 in paper [1], which shows the joint posterior distribution of the Hubble constant H_0 and inclination angles $\cos\iota$, using publicly available BNS merger data from the LIGO Document Control Center (DCC) and finding a value for H_0 through Bayesian statistical inference.

The report is structured as follows. First, the physical background is introduced. Then, there is a short explanation of the data acquisition followed by a description of the model. Next, the statistical framework and the inference procedure is explained. Subsequently, the results are presented and discussed. Finally, the report concludes with a summary of the key findings.

2 Physical Background

The Hubble constant H_0 which describes the mean expansion rate of the universe can be approximated as

$$v_H = H_0 d \tag{1}$$

for relatively short distances d ($d \lesssim 50Mpc$) between the source and detectors, where v_H is the velocity of the source solely due to the expansion of the universe named the local Hubble flow velocity [1]. The measured velocity, called the recession velocity, of the host galaxy NGC 4993 is inferred from its redshift [1]. To find the Hubble flow velocity v_H of the host galaxy the peculiar velocity, the velocity due to local gravitational pulls, v_p must be subtracted from the recession velocity [1]. Gravitational-waves emitted from the BNS merger are considered standard sirens as they give a direct measurement of the distance to the source [1] [8]. The luminosity distance d_L and inclination angle ι are inferred using Bayesian statistical methods from the observed gravitational wave strain, which arises from minute varying phase difference between laser light propagation in the detectors caused by passing gravitational waves [9]. Furthermore, there

is a correlation between the distance d_L and the inclination angle ι leading to a degeneracy between the two, contributing to the error in the distance which is inversely proportional to H_0 [1]. Next, a statistical Bayesian framework is used to infer the Hubble constant H_0 from the velocity, distance and inclination angle [1].

3 Data

The data was obtained from the public Ligo DCC. The file used is $GW170817_GWTC-1.hdf5$, and contains four datasets, the high-and low spin posteriors and priors. The low-spin datasets are based on a low-spin prior where the dimensionless spin of the neutron stars is restricted to $|\chi| \leq 0.05$ [9]. Whereas the high-spin datasets are based on a high-spin prior which allows for a broader spin range $|\chi| \leq 0.89$ [9]. However, the luminosity distance and inclination angle posteriors show negligible differences depending on the choice of prior, thus the high-spin posterior dataset was used for this project [9]. From the high-spin posterior dataset the luminosity distances and inclination angles are extracted using the python pandas library.

4 Model

Following *The 6dF Galaxy Survey: peculiar velocity field and cosmography* [10], a Bayesian approach is used to obtain peculiar velocity probability distributions for galaxies with $z < 0.055$, which applies for NGC4993. We apply a velocity correction for the peculiar velocity of the neutron stars, mirroring their approximations, as it is in the same velocity range. The NGC4993 neutron star event is not moving directly away or toward the Earth. Depending on its direction relative to us, the redshift requires a correction, typically modeled as a Gaussian uncertainty according to:

$$\sigma_{z,true} = \sqrt{\left(\frac{\sigma_{\text{pec vel}}}{c}\right)^2 + \sigma^2_{\text{red uncert}}}, \tag{2}$$

where $\sigma_{\text{pec vel}} = 250$ km s^{-1} is the peculiar velocity correction and $\sigma_{\text{red uncert}} = 1.67 \cdot 10^{-5}$, and $\sigma_{\text{pec vel}}$ has been divided by c to convert it from a velocity value to a redshift value following the definition $z \equiv v/c$ [11] [12].

5 Statistical Method

In this paper, the data was analyzed using a Bayesian framework whose goal is to update one's initial belief, portrayed as the prior, about the model M and its parameters $\boldsymbol{\theta}$ by evaluating the updated belief, represented by the posterior probability, given the observed data D and the prior information I as described in the following equation:

$$p(\boldsymbol{\theta}|M, D, I) = \frac{p(D, I|M, \boldsymbol{\theta})p(\boldsymbol{\theta}|M, I)}{p(D|M, I)}, \tag{3}$$

where $p(\boldsymbol{\theta}|M, D, I)$ is the posterior probability, $p(D, I|M, \boldsymbol{\theta})$ is the likelihood which measures how probable the observed data is given the model parameters, $p(\boldsymbol{\theta}|M, I)$ is the prior and $p(D|M, I)$ is the evidence, the latter being a normalization constant independent of the parameters and can therefore be omitted [13]. The core idea of the Bayesian approach resides in the fact that probabilities can be assigned to D as well as M and $\boldsymbol{\theta}$ [14]. Additionally, in a Bayesian framework, one refers to credible regions when considering the interval that contains $X\%$ of the posterior probability[14].

Therefore, one must first determine the likelihood and prior functions, as described in the following subsections, to obtain insight into the posterior probability and thus the model parameters.

After completing the inference procedure, explained in the next section, the posterior density function is plotted in a $\cos\iota$ vs H_0 graph which includes the most probable H_0 value - Maximum A Posteriori (MAP) - and the 68% credible region around it along with the SH0ES and Planck measured values of 73.30 ± 1.04 km s^{-1} Mpc^{-1} [7] and 67.4 ± 0.5 km s^{-1} Mpc^{-1} [6] respectively as references to be able to study the level of agreement.

The MAP is obtained using

$$\theta_{MAP} = \arg\max(p(\boldsymbol{\theta}|M, D, I)), \tag{4}$$

where the function $\arg\max(p(\boldsymbol{\theta}|M, D, I))$ returns the parameter value at which the posterior is maximized [14]. The 68% credible region was computed using the highest probability density (HPD) method, taking the integral downwards from the MAP that encloses the desired percentage of the posterior pdf [14]. In the code this is implemented by finding and selecting the narrowest possible interval containing 68% of samples. This is the HPD interval of 68% of the data. It is important to note that if the posterior has multiple peaks, this approach to calculating the HPD breaks down, as the true HPD may be multiple disjoint regions. However, from observing the produced 2D posterior graph it is trivial to see that the posterior of the data analysed here is not multimodal, making this method perfectly suitable.

5.1 Likelihood

In order to compute the H_0-$\cos\iota$ joint likelihood distribution over the data, one must start by estimating the joint posterior probability density function, $p(d_L, \cos\iota|GW)$, from the GW posterior samples; where d_L is the luminosity distance and $\cos\iota$ is the inclination angle[1]. As a means to deal with the two-dimensional nature of the pdf, a meshgrid approach was taken. The grid was constructed over H_0 and $\cos\iota$ and evaluated at a low redshift, described by $d_L = \frac{c}{H_0}z$, by applying a Gaussian kernel density estimate function, KDE, to it. This caused each grid point to have its own Gaussian distribution whose spread depends on the kernel bandwidth[13]; thus resulting in an overall smooth continuous surface that represents $p(dL, \cos\iota|GW)$. Furthermore, due to the fact that the prior is flat on $\cos\iota$, the likelihood is taken to be proportional to the posterior pdf [1] which is in accordance with Equation 3. Therefore, one can obtain the likelihood $p(GW|H_0, \cos\iota)$ by converting the luminosity distance into the Hubble constant which was achieved by multiplying the posterior pdf by the Jacobian given in Equation 5[15]. Finally, the logarithmic form of the likelihood was ultimately chosen due to not only its small values but also as a means to simplify the arithmetic computations in equation 3.

$$J = \left|\frac{\delta d_L}{\delta H_0}\right| = \left|-\frac{cz}{{H_0}^2}\right|, \tag{5}$$

where $\frac{\delta d_L}{\delta H_0}$ is the partial derivative of dL with respect to H_0 that was obtained by differentiating the low redshift approximation equation.

5.2 Priors

The two priors extracted from the reference paper [1] were a flat prior on $\cos\iota$ of $-1 < \cos\iota < 1$ and an flat-in-log prior of $\frac{1}{H_0}$ where $50 < H_0 < 100$. This leads to a log-prior function that takes position $(H_0, \cos\iota)$ and determines if it lies within the mentioned intervals and returns $-\log(H_0)$; or else it returns $-\infty$.

6 Inference Procedure

The goal of this section is to implement the Markov Chain Monte Carlo (MCMC) technique in order to infer the H_0 and $\cos\iota$ joint posterior probability and hence their expected values. This method combines two main ideas: the Markov Chain which is a sequence of random variables where the parameter θ solely depends on the preceding value; and the Monte Carlo integration which is a numerical method of estimating integrals, according to equation below, in which the random samples are drawn over the entire region[14].

$$\int g(\theta)p(\theta)d\theta \approx \frac{1}{N}\sum_{i}^{N} g(\theta_i), \tag{6}$$

where $p(\theta)$ is a shortened notation for the posterior pdf, $g(\theta)$ is the function that is being evaluated and $N = 100,000$ is the number of samples used[14].

The MCMC method consists of starting in an initial position, in this case at $(H_0 = 70, \cos\iota = -1)$ which is within the region with highest probability as shown in Figure 1; from which a Markov chain is constructed. Once the chain converges to a stationary distribution, one can obtain $p(\theta)$ since these two distributions are proportional to each other[14]. This behaviour occurs when satisfying the principle of detailed balance which states that the transition probability, $T(\theta_{i+1}|\theta_i)$, to go from one point to another is the same regardless of the direction of the jump, equivalently shown in the equation below[14].

$$T(\theta_{i+1}|\theta_i)p(\theta_i) = T(\theta_i|\theta_{i+1})p(\theta_{i+1}) \tag{7}$$

In practice, this was accomplished by implementing the Metropolis-Hastings Algorithm. The first step was to compute the proposal position from $T(\theta_{i+1}|\theta_i)$ which was taken to be a normal distribution $\mathcal{N}(\mu, \sigma)$ with $\mu = (H_{0,\text{current}}, \cos\iota_{\text{current}})$ and $\sigma = (w_{H_0} = 5, w_{\cos\iota} = 0.1)$ (the selection of these ω proposal widths will be revisited in the end of this section). This is symmetric in nature hence ensuring that Equation 7 is satisfied [14]. Next, the log-posterior was obtained for both the current and proposal positions by summing the corresponding log-likelihood and log-prior in accordance with the logarithmic form of Equation 3. Then, these posteriors pdfs were used to compute the log of the acceptance probability, p_{acc}, given by equation below[14].

$$\log\left[p_{acc}(\theta_i, \theta_{i+1})\right] = \log\left[p(\theta_{i+1})\right] - \log\left[p(\theta_i)\right] \tag{8}$$

The following step was to compare whether the logarithm of an arbitrary value from a uniform distribution was smaller than $\log\left[p_{acc}(\theta_i, \theta_{i+1})\right]$, if that was the case the log-posterior proposal was accepted and the corresponding H_0 and $\cos\iota$ values added to the chain; else the ones corresponding to the current log-posterior were added instead[14]. This ensures that the H_0 and $\cos\iota$ values are from samples that were drawn from high probability regions[14]. After running through all the samples, the created algorithm function, labeled as *Sampler 2D*, returned the acceptance probability as well as the accepted values. Finally, the Hubble constant was taken to be the MAP value given by Equation 4. The H_0 uncertainty computation is explained in the Appendix A.3.

As previously mentioned, one of the main requirements of this process is that the Markov chain reaches equilibrium[14]; practically this meant choosing the appropriate proposal width, w. This was done by studying the individual traces of H_0 and $\cos\iota$ as well as their combined trace for different proposal widths. In the end, the case which met with the two conditions - of exploring the majority of the parameter space and having $p_{acc} \approx 20\%$ [16] - was selected, refer to Figure 5 located at the Appendix A.4. Furthermore, a burn-in of 100 - an approach

which neglects the first X initial accepted values where the chain hasn't yet stabilized [14]- was implemented.

7 Results

7.1 Parameter Constraints

Parameter	Estimate (km s^{-1} Mpc^{-1})	Uncertainty
H_0	68.26	$^{+12.85}_{-5.21}$

Table 1: Posterior parameter estimate for Hubble's constant H_0.

Table 1 shows the posterior parameter for Hubble's constant which is $H_0 = 68.26^{+12.85}_{-5.21}$ km s^{-1} Mpc^{-1}

7.2 Model Comparison With Data

Figure 1 shows a two dimensional histogram of the raw data to help visualize it.

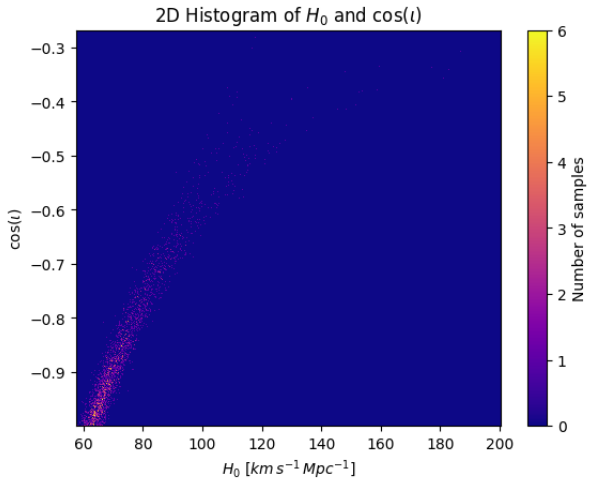
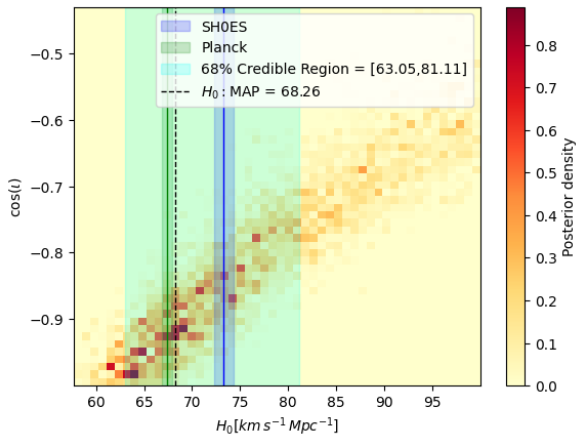
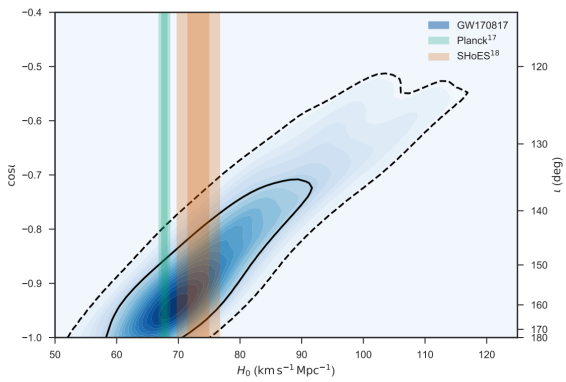


Figure 1: A 2D histogram of the raw data of H_0 and $\cos\iota$.

Figure 2 shows a comparison between the replication of Figure 2 from paper [1] in Figure 2a and the original graph presented in Figure 2b. Both graphs in Figure 2a and Figure 2b have the same overall shape. Furthermore, the value of the Hubble constant found in this report is $H_0 = 68^{+13}_{-6}$ km s^{-1} Mpc^{-1} which lies within the uncertainties of $H_0 = 70^{+12.0}_{-8.0}$ km s^{-1} Mpc^{-1} determined in paper [1].



(a) Inference on H_0 and $\cos\iota$ with bandwidth 1, where the dashed line and light blue region represent the estimated MAP and the 68% credible interval respectively; and the dark green and blue solid lines represent the Planck and SH0ES measurements respectively with their uncertainty given by the corresponding shaded regions.



(b) Figure 2 from paper [1] showing the joint posterior density of H_0 and $\cos\iota$. The green and red lines represent the Planck and SH0ES measurements of H_0 respectively with 1- and 2σ error bands. The solid and shaded black line show the 1σ and 2σ credible levels respectively.

Figure 2: Comparison of the reproduced figure and original Figure 2 in paper [1]. The inferred value of $H_0 = 68^{+13}_{-6}$ km s^{-1} Mpc^{-1} found in this report is within the uncertainty range of the value found in paper [1] which is $H_0 = 70^{+12.0}_{-8.0}$ km s^{-1} Mpc^{-1}.

8 Discussion

The Hubble constant found through MAP is $H_0 = 68^{+13}_{-5}$ km s^{-1} Mpc^{-1}. MAP was used as it is part of the Bayesian statistical framework. The uncertainty in the value is large as the data used are posterior distribution inferred from a single measurement [4]. Additionally, the luminosity distance d_L and inclination angle ι are correlated resulting in a degeneracy between the two, contributing to the uncertainty in H_0 as from Equation 1 $H_0 \propto \frac{1}{d_L}$ [1]. The estimated value found is consistent with both the Planck ($H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1}) and SH0ES (73.30 ± 1.04 km s^{-1} Mpc^{-1}) values of H_0 and does not help resolve the Hubble tension [6] [7]. Furthermore, it is within the uncertainty range of the value found in paper [1] which is $H_0 = 70^{+12.0}_{-8.0}$ km s^{-1} Mpc^{-1}.

One of the first steps in this process is to obtain the likelihood by employing the KDE which is rather sensitive to the kernel's bandwidth. This can be seen when comparing the likelihood surface plots for the bandwidths of $0.1, 1$ and 10 as shown in Figure 3 respectively (refer to Appendix A.1); where the plot on the far left is overfitted causing the model to be too sensitive to the data points thus producing noise; the middle plot shows the best balance between smoothness and retained information therefore being the ideal choice; and the far right plot is underfitted leading to a loss of information. Furthermore, the plotted credible regions scale with the bandwidth as shown in Figures 2a, 6 and 7 located in the Appendix A.5. The credible region found for bandwidth 1 is $[63.05, 81.11]$.

Now considering MCMC, from Figure 1 it is clear that a suitable initial guess for the MCMC should lie within $60 \lesssim H_0 \lesssim 80$ and $-1 < \cos\iota \lesssim -0.8$, hence the initial values are chosen to be $H_0 = 70$ and $\cos\iota = -1$. In order to find an appropriate proposal width for the MCMC, trace plots for H_0 and $\cos\iota$ and a combined trace plot are created for a range of different widths (refer to the notebook in Appendix C.3), Figure 5 shows the trace plots for proposal widths

$w_{H_0} = 5$ and $w_{cos\iota} = 0.01$ these were found to be the most apt values as the MCMC chain explores almost the entirety of the parameter space and has an acceptance rate of 23.9% [16]. However, the MCMC struggles to explore the space for high values of H_0 and $\cos\iota$ because of the strong localization of the likelihood. Added to this, the individual trace plots of H_0 and $\cos\iota$ do not seem to converge. Moreover, a burn-in value of 100 is used which is relatively low as Figure 1 gives an appropriate initial guess, additionally larger burn-in values do not change the final result.

Figure 2 shows a comparison of the reproduced and original Figure 2 in paper [1]. Figure 2a and Figure 2b illustrate the joint posterior pdf of H_0 and $\cos\iota$, both show the same overall trend. Moreover, these are plotted using the bandwidths of 0.1 resulting in $H_0 = 66^{+9}_{-5}\ km\,s^{-1}\,Mpc^{-1}$ and 10 resulting in $H_0 = 64^{+19}_{-7}\ km\,s^{-1}\,Mpc^{-1}$ which produce Figures 6 and 7. From these it is clear that regardless of the bandwidth used, the Planck and SH0ES H_0 values are contained within the credible region which shows the robustness of the model used. Nevertheless, using bandwidth 1 provides a more precise result for H_0.

9 Possible Extensions

The main improvement to the H_0 result using the Standard Siren method would be a larger data set. However, it is uncommon to observe both gravitational wave and electromagnetic wave signals needed for this approach. As explained, the data used was therefore made up from posterior samples from a single measurement, which has as a consequence that the uncertainties are large. This is of great importance if the H_0 values obtained through this method are to add to the conversation regarding the Hubble tension.
Additionally, the influence of using different priors on the results could be investigated. For instance the obtained results could be compared to those obtained by not setting bounds on the H_0 prior. This would be interesting to determine whether the results are data- or prior-dominated.
Furthermore, the python package emcee could be used as it yields a higher acceptance rate and a more reliable appearing trace, as discussed in Appendix B.
Finally, another improvement would be a more rigorous approach to determining an appropriate proposal width. This could be done by plotting histograms of the obtained H_0 values in the MCMC chain through the steps. This would help visualise whether the value converges to the expected distribution, which in turn could be use to discard an appropriate burn-in. The proposal widths could also be determined dynamically, using an Adaptive Metropolis-Hastings algorithm who's chain self-optimises by selecting proposal widths as a function of the current covariance matrix of the posterior built by the current chain history.

10 Conclusion

The introduction of the gravitational-wave standard siren measurement to evaluate the Hubble constant is used in the hopes of resolving the Hubble tension. In order to determine H_0 a Bayesian statistical framework is used. First, the joint posterior likelihood of H_0 and $\cos\iota$ is obtained through the gaussian KDE method. Then, MCMC using the Metropolis-Hastings algorithm with a flat-log prior on H_0 and a flat prior on $\cos\iota$ is applied to infer the joint posterior density function. Finally, the credible region of H_0 which is $[63.05, 81.11]$ is determined and MAP is used to obtain the value of $H_0 = 68^{+13}_{-6} kms^{-1}Mpc^{-1}$. The reproduced figure strongly resembles the original Figure 2 in paper [1]. Furthermore, the value inferred for H_0 lies within the results of Planck, SH0ES and that found in paper [1] showing the robustness of the model used [6] [7]. Therefore, the result of H_0 found in this report does not aid in solving the Hubble tension.

Author Contributions

- Elisa: **Code:** likelihood function with KDE implementation, redshift error propagation, proposal widths determination using trace and acceptance rates, effect of KDE bandwidths on results, notebook markdown cell write-ups; **Report:** Abstract, Possible Extensions
- Leticia: **Report:** Statistical Method, Inference Procedure, Results, Discussion, Conclusion, Additional Figures
- Michael: **Code:** MCMC algorithm implementation, log likelihood function, MAP and HPD function, plot generation, emcee implementation, redshift error propagation, repository management, notebook markdown cell write-ups; **Report:** Statistical Method, Model, Discussion, Reproducibility, emcee Comparison
- Horia: Model
- Elliot: **Code:** Data Acquisition; **Report:** Introduction, Physical Background, Data, Results, Discussion, Conclusion

References

[1] B P Abbott et al. "A gravitational-wave standard siren measurement of the Hubble constant". In: *Nature* 551.7678 (2017), pp. 85–88. ISSN: 1476-4687. DOI: 10.1038/nature24471. URL: https://doi.org/10.1038/nature24471.

[2] Eleonora Di Valentino and Dillon Brout. *The Hubble constant tension*. Ed. by Eleonora Di Valentino and Dillon Brout. Springer Series in Astrophysics and Cosmology. Singapore: Singapore : Springer, [2024], 2024. Chap. Chapter 35, pp. 661–674. ISBN: 9789819901777. DOI: 10.1007/978-981-99-0177-7. URL: https://link.springer.com/10.1007/978-981-99-0177-7.

[3] Edwin Hubble. "A relation between distance and radial velocity among extra-galactic nebulae". In: *Proceedings of the National Academy of Sciences* 15.3 (Mar. 1929), pp. 168–173. ISSN: 0027-8424. DOI: 10.1073/PNAS.15.3.168. URL: https://www.pnas.org/doi/abs/10.1073/pnas.15.3.168.

[4] Elcio Abdalla et al. *Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies*. June 2022. DOI: 10.1016/j.jheap.2022.04.002.

[5] Licia Verde, Tommaso Treu, and Adam G. Riess. "Tensions between the early and late Universe". In: *Nature Astronomy 2019 3:10* 3.10 (Sept. 2019), pp. 891–895. ISSN: 2397-3366. DOI: 10.1038/s41550-019-0902-0. URL: https://www.nature.com/articles/s41550-019-0902-0.

[6] N. Aghanim et al. "Planck 2018 results. VI. Cosmological parameters". In: *Astronomy and Astrophysics* 641 (July 2018). ISSN: 14320746. DOI: 10.1051/0004-6361/201833910. URL: https://arxiv.org/abs/1807.06209v4.

[7] Adam G. Riess et al. "A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km s1 Mpc1 Uncertainty from the Hubble Space Telescope and the SH0ES Team". In: *The Astrophysical Journal Letters* 934.1 (July 2022), p. L7. ISSN: 2041-8205. DOI: 10.3847/2041-8213/AC5C5B. URL: https://doi.org/10.48550/arXiv.2112.04510.

[8] Hsin-Yu Chen et al. "A standard siren cosmological measurement from the potential GW190521 electromagnetic counterpart ZTF19abanrhr". In: *MNRAS* 513 (2022), pp. 2152–2157. DOI: 10.1093/mnras/stac989. URL: https://doi.org/10.1093/mnras/stac989.

[9] B. P. Abbott et al. "GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral". In: *Physical Review Letters* 119.16 (Oct. 2017), p. 161101. ISSN: 10797114. DOI: 10.1103/PHYSREVLETT.119.161101/DELIVERABLE/24890079-4049-42AC-9C0D-E9CC23B11D98. URL: https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.119.161101.

[10] Christopher M. Springob et al. "The 6dF Galaxy Survey: peculiar velocity field and cosmography". In: *Monthly Notices of the Royal Astronomical Society* 445.3 (Dec. 2014), pp. 2677–2697. ISSN: 0035-8711. DOI: 10.1093/MNRAS/STU1743. URL: https://dx.doi.org/10.1093/mnras/stu1743.

[11] D. Brout et al. "First Cosmology Results Using SNe Ia from the Dark Energy Survey: Analysis, Systematic Uncertainties, and Validation". In: *The Astrophysical Journal* 874.2 (Apr. 2019), p. 150. ISSN: 0004-637X. DOI: 10.3847/1538-4357/AB08A0. URL: https://iopscience.iop.org/article/10.3847/1538-4357/ab08a0%20https://iopscience.iop.org/article/10.3847/1538-4357/ab08a0/meta.

[12] *By Name — NASA/IPAC Extragalactic Database*. URL: https://ned.ipac.caltech.edu/byname?objname=ngc+4993&hconst=67.8&omegam=0.308&omegav=0.692&wmap=4&corr_z=1.

[13] Željko Ivezić et al. "Classical Statistical Inference". In: *Statistics, Data Mining, and Machine Learning in Astronomy Book Subtitle: A Practical Python Guide for the Analysis of Survey Data*. Updated Edition. Princeton University Press, 2020. Chap. Chapter 4, pp. 115–164. DOI: https://doi-org.proxy-ub.rug.nl/10.2307/j.ctvrxk1hs.7. URL: https://about.jstor.org/terms.

[14] Željko Ivezić et al. "Bayesian Statistical Inference". In: *Statistics, Data Mining, and Machine Learning in Astronomy Book Subtitle: A Practical Python Guide for the Analysis of Survey Data*. Updated Edition. Princeton University Press, 2020. Chap. Chapter 5, pp. 165–240. ISBN: 202619:34:28. DOI: https://doi-org.proxy-ub.rug.nl/10.2307/j.ctvrxk1hs.8. URL: http://www.jstor.com/stable/j.ctvrxk1hs.8.

[15] Watson Saleem Stewart James Clegg Dan. "Multiple Integrals". In: *Calculus : early transcendentals*. Ninth edition. Singapore : Cengage, 2021. Chap. Chapter 15, pp. 1037–1122.

[16] Thibeau Wouters et al. "Robust parameter estimation within minutes on gravitational wave signals from binary neutron star inspirals". In: *Physical Review D* 110.8 (Oct. 2024). ISSN: 24700029. DOI: 10.1103/PHYSREVD.110.083033.

A Additional Figures & Error Propagation

A.1 Likelihood vs Bandwidth Plot

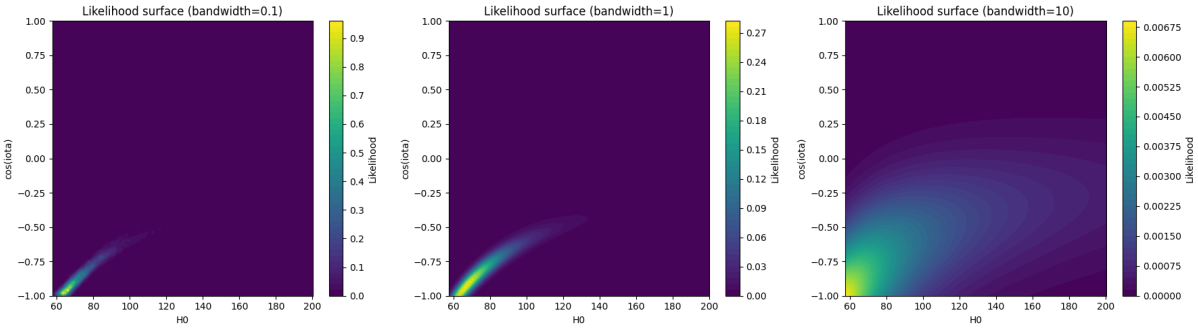


Figure 3: Likelihood surface plots for different kernel's bandwidth of $0.1, 1$ and 10 respectively.

A.2 Marginalised posteriors

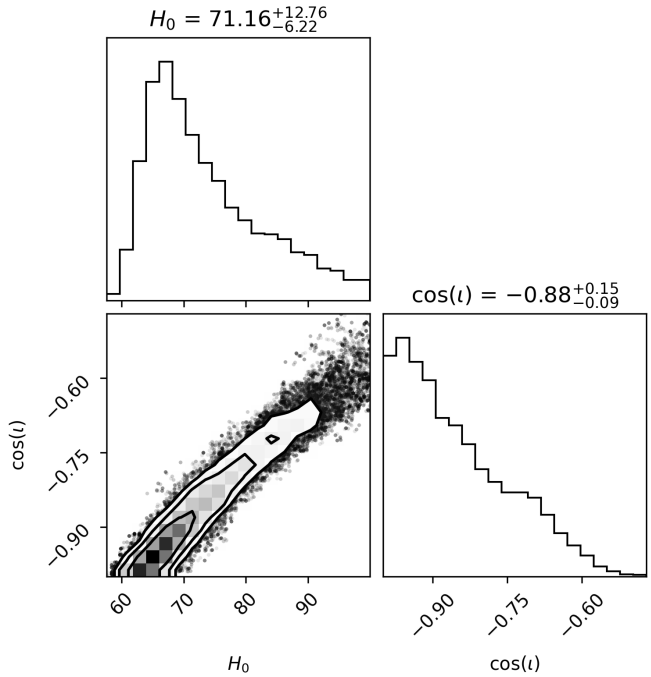


Figure 4: The 2D posterior plotted as a contour plot, and the marginalised 1-dimensional posteriors of H_0 and $\cos\iota$ using the corner package. Note that the values shown for H_0 and $\cos\iota$ in the titles are the medians and 68% credible region as defined by quantiles, not the MAP and HPD that have been used in this reports analysis and shown in Figure 2a.

A.3 Error Propagation

The redshift of the host galaxy in the CMB frame is used to correct for the motion of the observer. The remaining uncertainty from the host galaxy's peculiar velocity is modeled as a Gaussian with $\sigma_{\text{pec}} \approx 250, \text{km}, \text{s}^{-1}$, corresponding to a redshift uncertainty $\sigma_z = \sigma_{\text{pec}}/c$. This is added in quadrature to the measurement uncertainty and implemented into the likelihood using a Gaussian weighting, resulting in a "smearing" of the likelihood grid.

A.4 Trace Plot

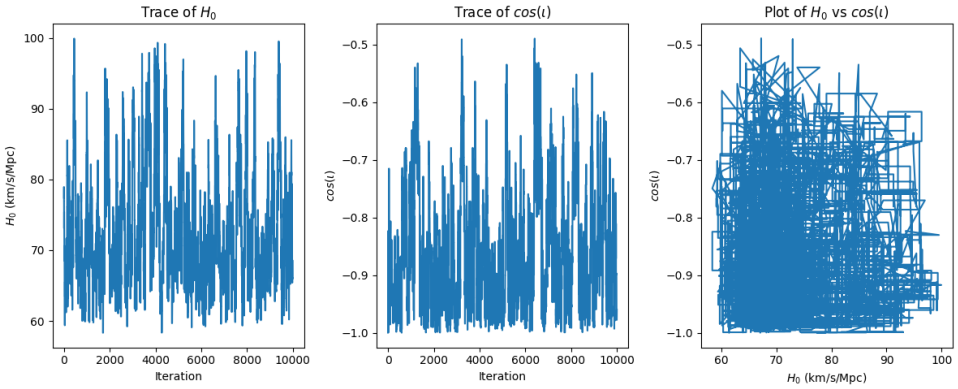


Figure 5: Trace plots for a proposal width of $(w_{H_0} = 5, w_{\cos\iota} = 0.01)$ with an acceptance rate of 23.9%.

A.5 Inference vs Bandwidth Plots

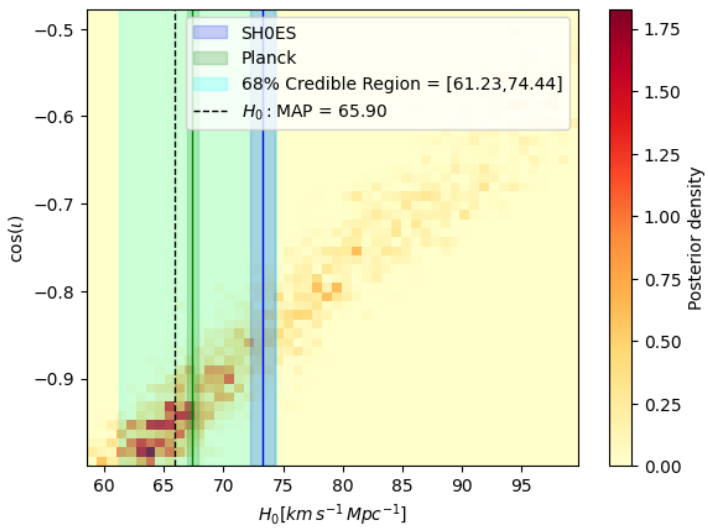


Figure 6: Inference on H_0 and $\cos\iota$ with bandwidth 0.1, where the dashed line and light blue region represent the MAP value and the 68% credible interval respectively; and the dark green and blue solid lines represent the Planck and SH0ES measurements respectively with their uncertainty given by the corresponding shaded regions. It was obtained that $H_0 = 66^{+9}_{-5}$ km s^{-1} Mpc^{-1}.

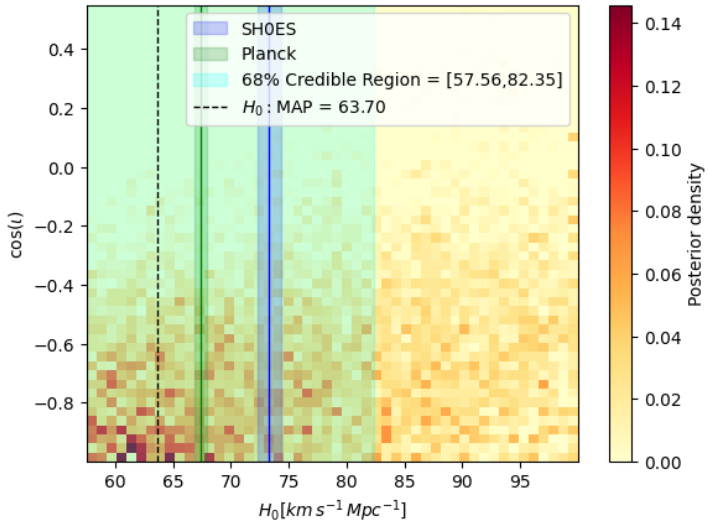


Figure 7: Inference on H_0 and $\cos\iota$ with bandwidth 10, where the dashed line and light blue region represent the MAP value and the 68% credible interval respectively; and the dark green and blue solid lines represent the Planck and SH0ES measurements respectively with their uncertainty given by the corresponding shaded regions. It was obtained that $H_0 = 64^{+19}_{-7}$ km s^{-1} Mpc^{-1}.

B emcee Comparison

To evaluate the performance of the in-house, single chain Metropolis-Hastings algorithm used, the same data was sampled using the well documented emcee python package and plots were produced. The values and posterior obtained by emcee show a high degree of agreement with posterior obtained by the in-house implementation, albeit with a much higher acceptance rate of 63.6% and a much more reliable appearing trace.

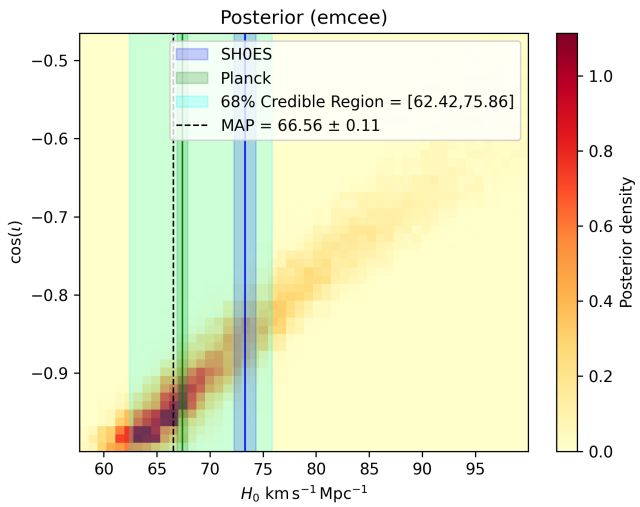


Figure 8: The main 2D inference posterior, obtained using the emcee package instead of the in-house developed Metropolis-Hastings algorithm.

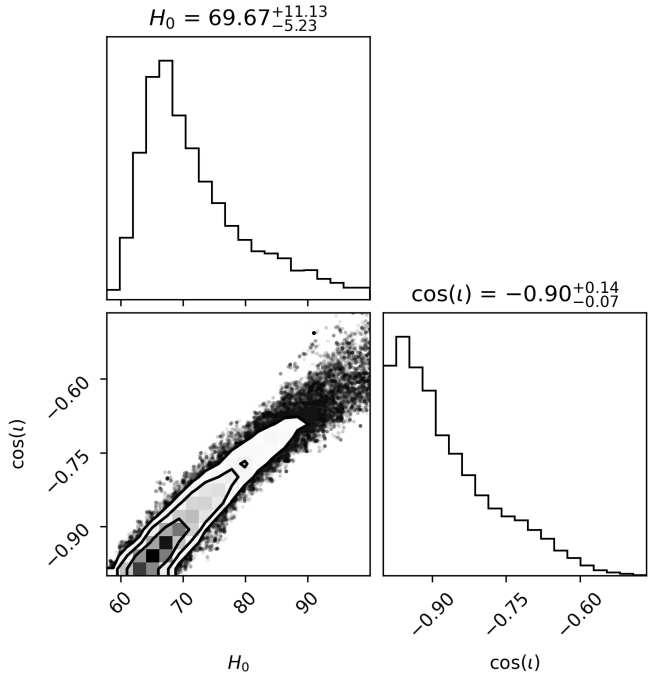


Figure 9: The corner plot, obtained using the emcee package instead of the in-house developed Metropolis-Hastings algorithm.

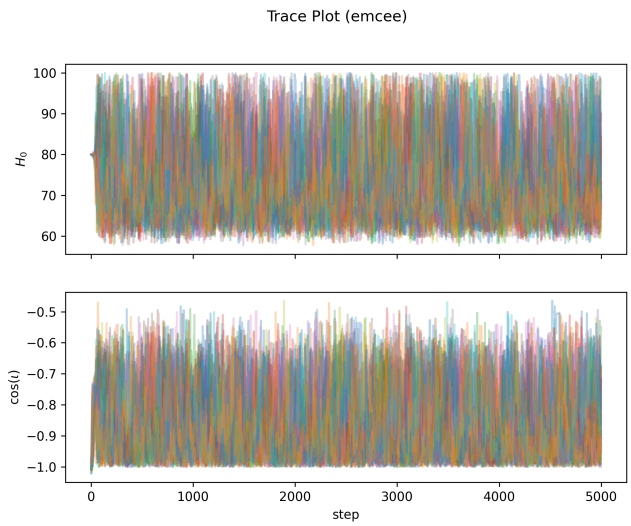


Figure 10: The traces of the 32 emcee walkers.

C Reproducibility

C.1 Software and Packages

The analysis was performed in Python. The following packages were used (versions in parentheses):

- corner (2.2.3)
- h5py (3.16.0)

- matplotlib (3.10.8)
- numpy (2.4.3)
- pandas (3.0.1)
- scipy (1.17.1)
- emcee (3.1.6)

C.2 Instructions for Running the Code

Download the GW170817 posterior samples from `https://dcc.ligo.org/LIGO-P1800370/public`, and place the downloaded file in the `data/` folder, under the name `GW170817_GWTC-1.hdf5`. To reproduce the main plot seen in Figure 2a, run

```
python src/GW_code.py
```

To reproduce the intermediary plots seen in the appendix, run

```
jupyter notebook notebooks/GW_notebook.ipynb
```

C.3 GitHub Repository

The public GitHub repository can be found at `https://github.com/mjpineda-rug/GW1_H0_Sirens.git`

D Feedback

We think this assignment is an effective way of applying our statistic knowledge in real problems. Also, we think it was of adequate difficulty therefore we give it an 8/10. However, the proposed division of labour was quite unbalanced, and lacked a role/roles that would work on the report. Furthermore and more importantly, we would have preferred if the deadline wasn't the weekend before exam season.